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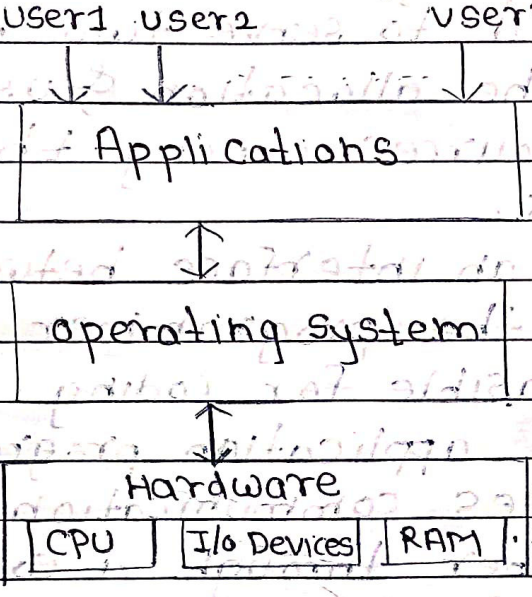
UNIT 1 - Introduction to operating system

Introduction to operating system

- An operating system (OS) is the foundational software that manages a computer's hardware & software resources, acting as an intermediatory between the user & computer



- The operating system as a user interface :
- User (Interacts with the system)
- System & application programs (Software tools for the user)
- operating system (Manages hardware & SW resources)
- Hardware (physical computing devices)
- Every general-purpose computer consists of hardware, an operating system (S), system program & application programs.

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- The hardware consists of memory, CPU, ALU, I/o devices, peripheral devices & storage devices.
- The System program consist of compilers, loders, editors, os, etc.
- Every computer must have an operating system to run other programs.
- It simply provides an environment within which other programs can do useful work.
- OS is designed to serve two basic purposes.
- It controls the allocation & use of computing System's resources among the various user of tasks.
- It provides an interface between the computer hardware & the programmer that simplifies & markes feasible for coding, creating, debugging of application programs.
- The OS handles communication between the computer, either through a command line its pheripherals (like keyboard, mouse, display etc)
- The operating System (OS) manages of all the software & hardware on the computer.
- operating systems can be broadly categorized into several types, including batch, time-Sharing, real-time distributed, and mobile operating system.
- An OS is a package of some programs that runs on a computer machine, allowing it to perform efficiently.

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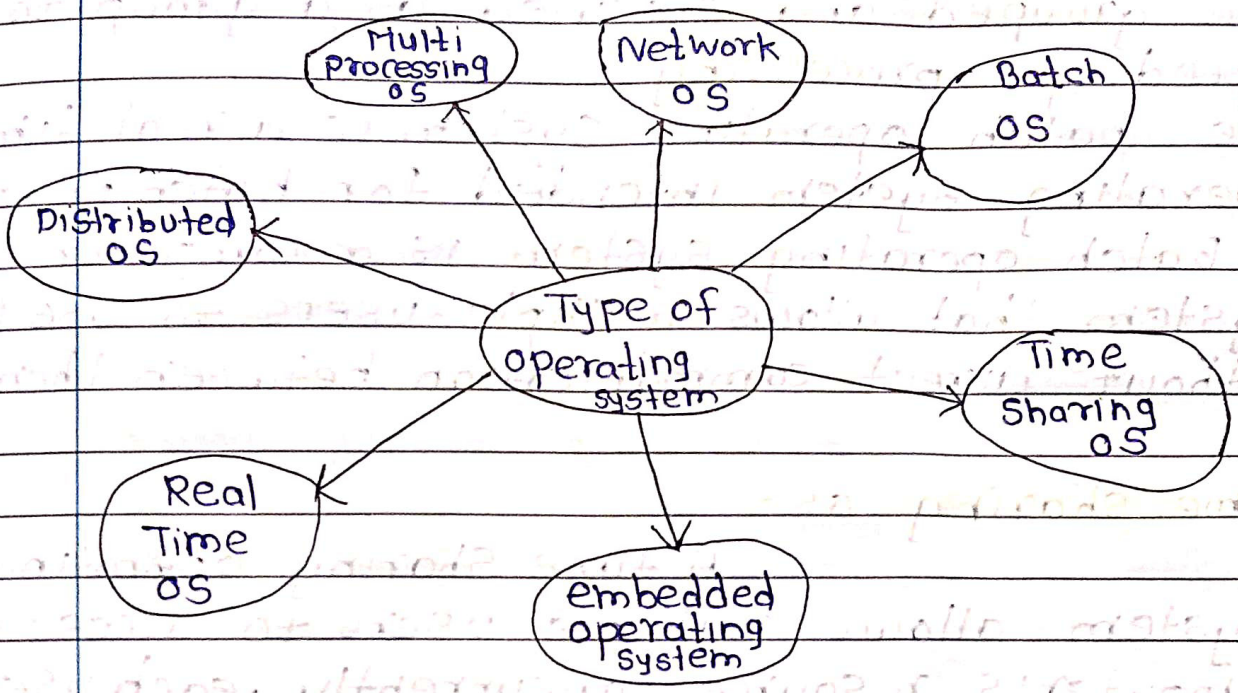
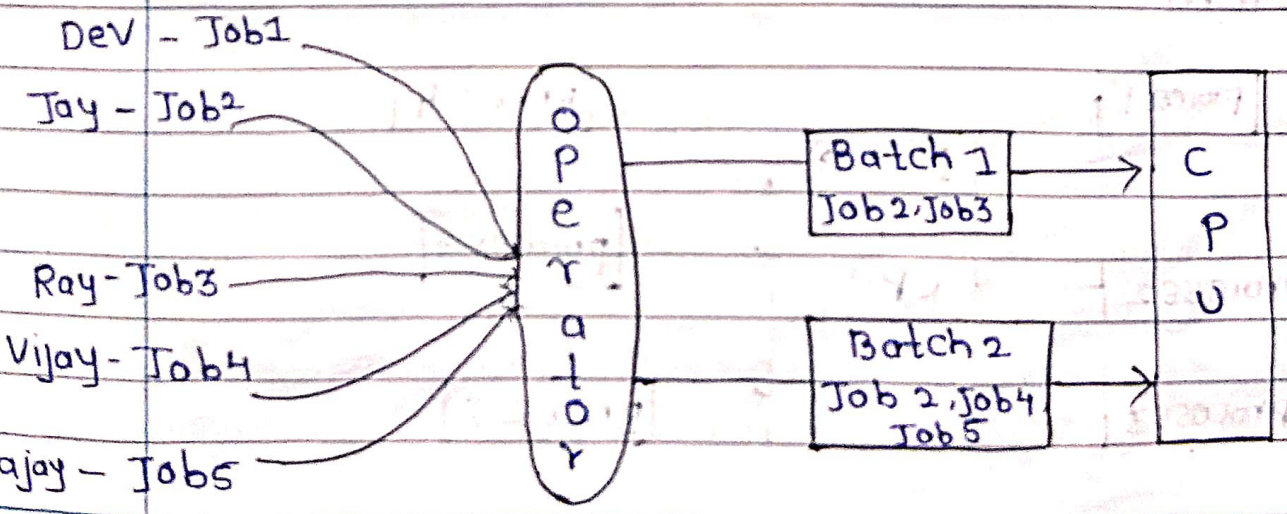
Type of operating system



- **Batch operating System**
  - Jobs are grouped into batches based on similar requirements and executed sequentially.
  - A batch operating system is a type of operating system that processes jobs in batches, without requiring immediate user interaction.
  - In batch operating system System the jobs where performed in batches.



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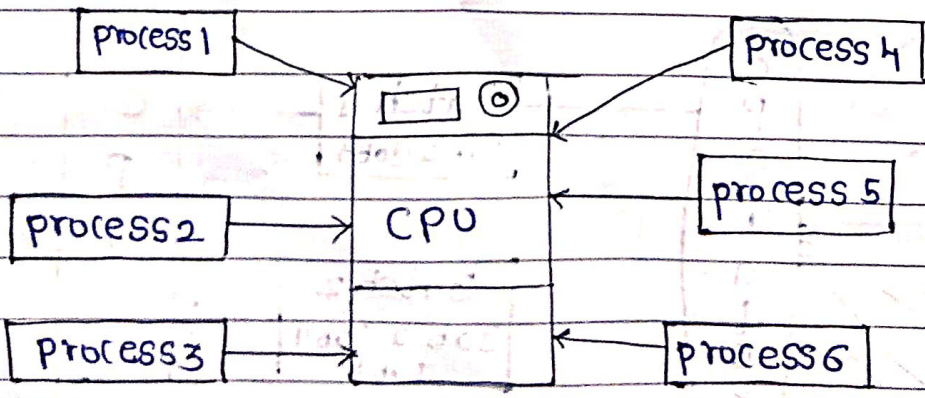
- This means jobs having similar requirements are grouped and executed as a group to speed up processing.
- The batch operating system is a real time operating system intended for batch processing
- A Batch operating system is a computer system that allows multiple users to use it without direct communication between them.

• Time Sharing os -

A time sharing operating system allows multiple users to access a computer's resource concurrently, each user seemingly having their own dedicated system

- A time sharing operating system allows multiple user to use a single computer at the same time.
- Time-Sharing operating system is a type of operating system that enables multiple people to use to the same Computer simultaneously.
- Linux is an open-source time sharing operating system.



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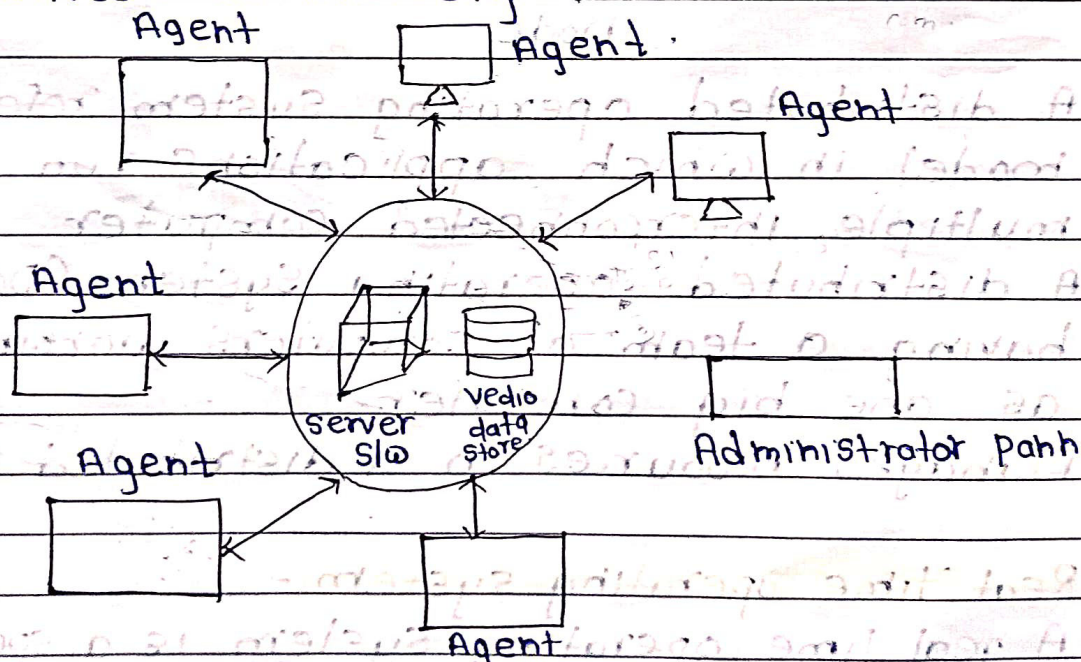
- **Network operating system-**

A network operating System (NOS) is specilized software that allows computers within a computer n/w to communicate & share resources likes. files, printers & internet access.

- Network Systems are the hardware & s/w components from which these networks are built.
- There are two primary parts of most network operating systems:
  Client
  Server
- NOS is a software designed to manage n/w resources effectively



- **Distributed operating system (DOS)-**

A distributed operating System (DOS) manages a network of independent computers, making them appear as a single

user & applications.

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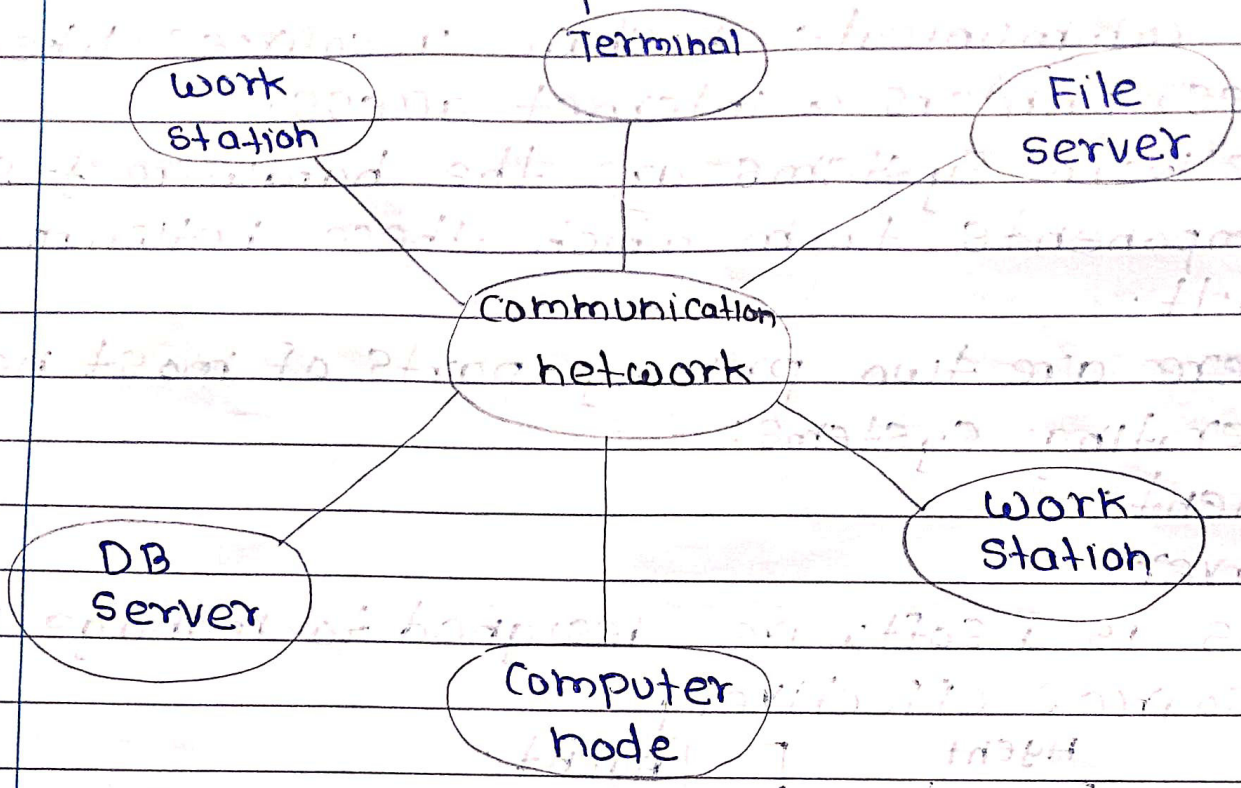
- A distributed OS is like having multiple computers working together as one instead of just one powerful computer



- A distributed operating system refers to a model in which applications run on multiple interconnected computer
- A distributed operating system (DOS) is like having a team of computers working together as one big computer.
- Manages resources in a distributed system.

Real Time operating system.

- A real Time operating system is a specialized type of operating system designed for applications where timing is critical. unlike regular operating system that might delay tasks, an RTOS that task are

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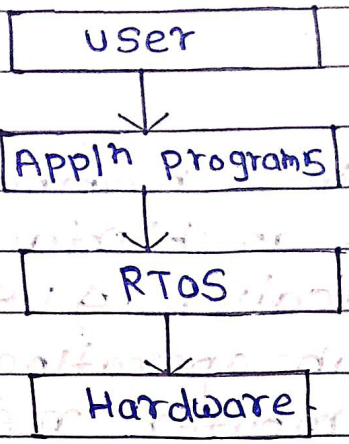

Completed with strict deadline.

- Real time refers to immediate data processing without delays.
- Common examples of real time systems include air traffic control system, process control system.

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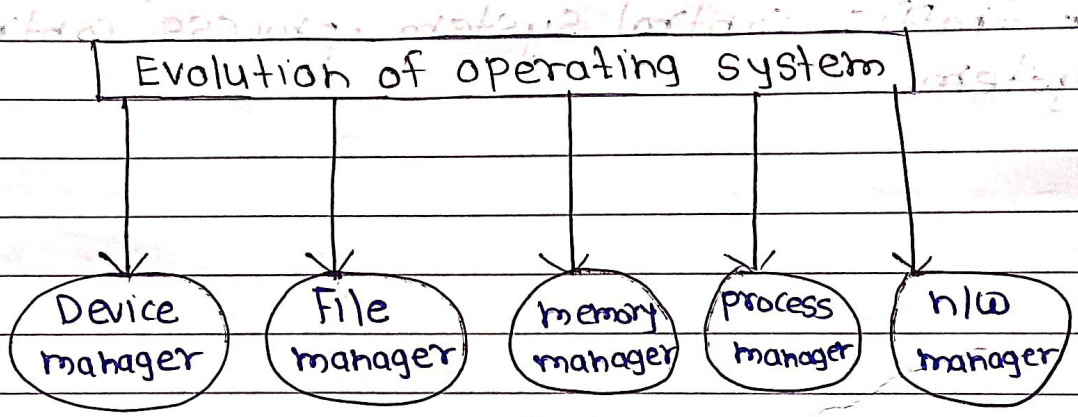
Evolution of OS -

- The operating system is nothing but a s/w program that act as an interface between the hardware, the application, & user.
- The main aim of an operating system is to manage all the computer resources, Security, File systems etc.
- So we can simply say that operating system gives a platform to the application s/w & other system s/w to perform their task.



- The overall evolution period of the computer system are divided into different generation & have provided machines according to the needs & types of the users.
- This can be better described in tearms of examples including mainframe computers to today's hand held devices.
- In the evolution following are type of the operating System have been using since beginning of the computing period with the

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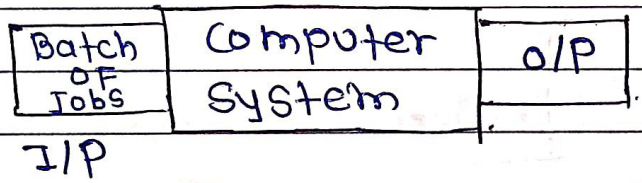
assistance of operating systems as managers.
- Simple Batch processing system.
- Multi programming Batch processing system
- Multi-process systems.
- Time sharing systems
- Multi-user system
- Real Time systems.
- In the early age of computing, the computer systems were very large & expensive & the need for enhanced processor utilization was required.



- The error situations are alert by some mechanisms sometimes by using red colour lights then after corrections again jobs are submitted in the form of batch of processing.
- Multiprogramming simply referred as the existence of multiple programs in the computer's memory at any instance of time in non-interactive environments.
- The overall time required for processing in the simple batch processing system includes jobs submission into the system, formation of batch of jobs & results of executions of these jobs.

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- **OS structure** : Layered Approach, Monolithic, Microkernel operating systems.

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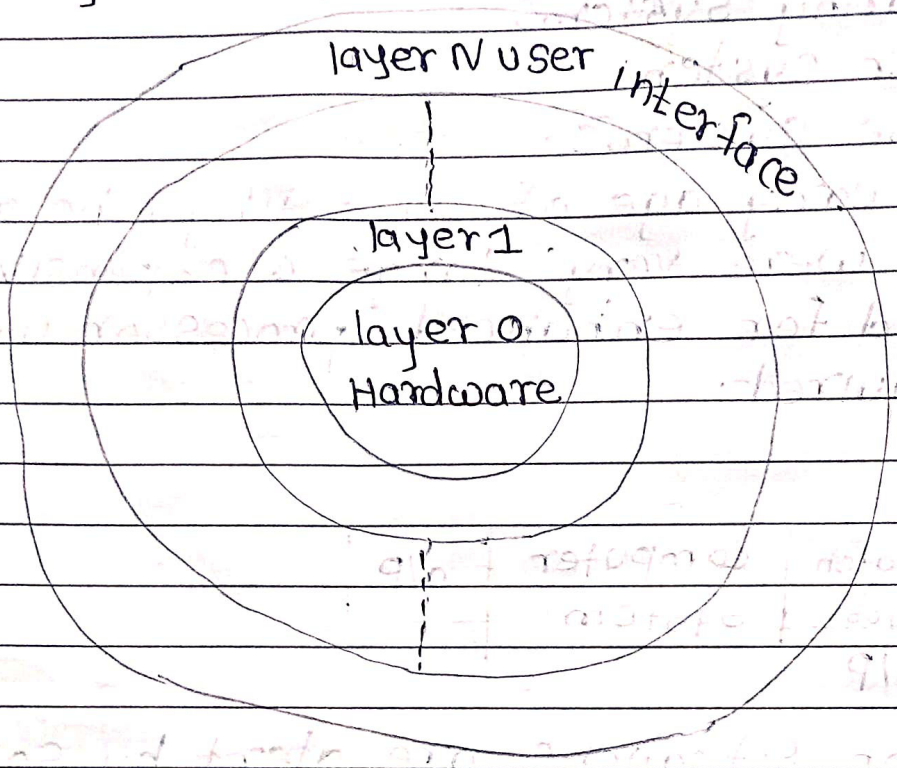
- **Layered Approach** -



- A layered approach is an operating system (OS) divides the OS into multiple layers, each performing specific functions.
- layered structure is the type of op system structure in which the different services of the operating system are split into various layers.
- As we know, there are many processes that a task undergoes before executing in a system. These processes are performed by different part of a system may be software, hardware etc.
- All these sections of processes are termed

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layers & this concept of arranging different tasks into various layers is called the layered structure of operating system.

- The bottom-most layer is the hardware & the top layer is the user interface.
- There are many things such as memory, S/W, microprocessor etc that combine to from a proper system.
- The construction & debugging of the layered structure are easy & simple.
- Each of the layers relies on a layer that is below it.
- The innermost layer should be the H/W layer.
- The outermost layer should be the user interface layer.
- layer 0 handles the basic multiprocessing of the CPU.

• Advantages of layered structure.

- Modularity
- Easy debugging
- Easy update.
- No direct access to H/W.
- Abstraction

• Disadvantages

- Complex & careful implementation
- Slower execution

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Monolithic OS structure -

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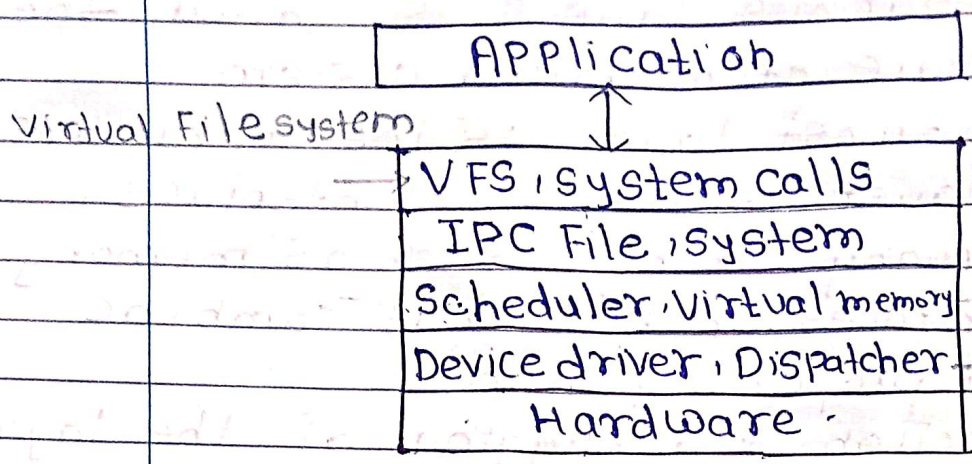

Monolithic OS kernel

- The monolithic operating system is a very simple operating system where the kernel directly controls device management, memory management, File management & process management.
- All of the system's resources are accessible to the kernel.
- Every part of operating system is contained within the kernel is monolithic system
- The monolithic kernel is another name for the monolithic operating system.
- This is outdated operating system that banks employ for job likes time-sharing & batch processing.
- All hardware components are managed by the monolithic kernel, which function as a virtual machine.
- Unlike a microkernel, which has fewer

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tasks, it is not the same.
- The two sections of a microkernel are called kernel space & user space.
- In a monolithic architecture, the operating system kernel is designed to provide all operating system services, including memory management, process scheduling, device drivers & file systems, in a single, large binary.
- The main advantages of monolithic architecture is that it can provide high performance since system calls can be made directly to the kernel without the overhead of message passing between user-level processes.
- Two basic instances of monolithic operating systems are DOS & CP/M.
- operating systems that share a single address space with applications are DOS & CP/M.
- A monolithic OS structure is characterized by its large, single-kernel that contains all core of OS functions & services.
- This structure is known for its speed & efficiency due to the lack of overhead from communication between different components.

Advantages	Disadvantage
• Simplicity of development	• Slow speed of development
• Simplicity of deployment	• Testing become harder
• Simplicity of testing	• performance issue
• Simplicity of debugging	

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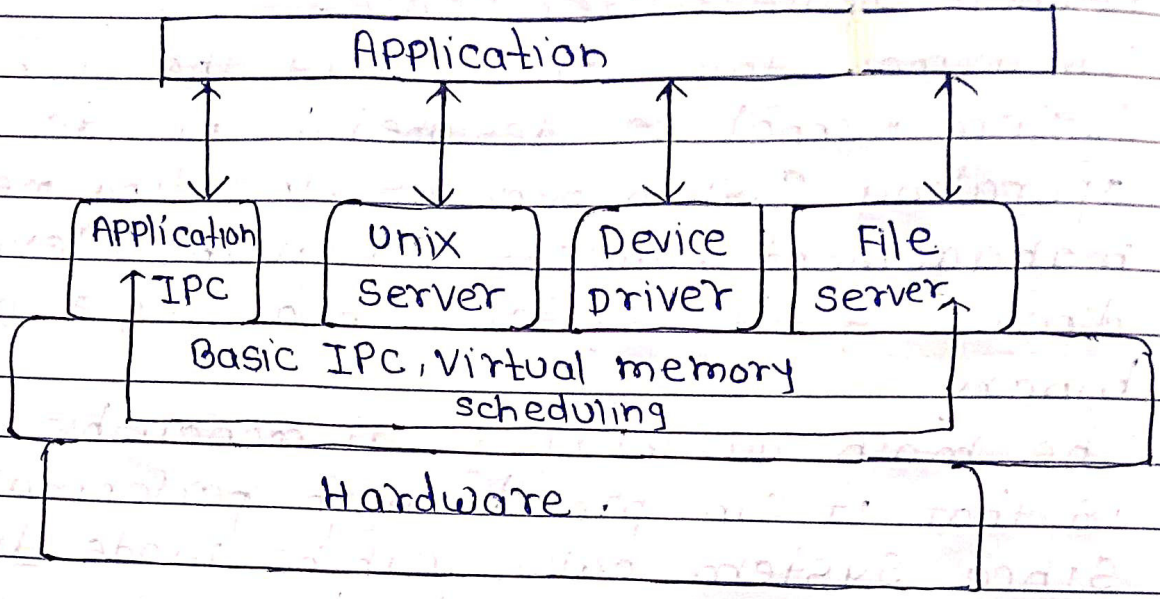
- Microkernel operating system (OS) structure



- A microkernel is an approach to designing an operating system.
- The microkernel provides very fundamental services required to run the OS like basic memory management, task scheduling etc.
- Microkernels use interprocess communication (IPC) for communication.
- A kernel is the core part of an operating system that manages system resource.
- It also acts as a bridge between the application & Hardware of the computer.
- It is one of the first programs loaded on start-up.
- Microkernel is a type of operating system that provides some basic services of an operating system.
- These services include memory management

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, process scheduling etc.
- Some other service like Device drivers, File systems etc are managed by user level processes.
- user level process communicates with microkernel via message passing.
- Microkernels provide a minimalist approach to operating system design by running only essential services in kernel space, making the OS more modular & secure.
- The operating system puts the CPU in kernel mode when it is executing in the kernel so, that kernel can execute some special operation.
- System calls are typically implemented in the form of software interrupts, which causes the hardware interrupt handler to transfer control over to an appropriate interrupt handler, which is the part of the operating system, switching the bit mode to kernel mode in the process.
- The important services for which the microkernel is responsible are
- Inter-process communication
- Scheduling of CPU
- Memory management
- Interprocess Communication it refers to how processes interact with each other.
- Scheduling of CPU it refer to which process will be executed next.

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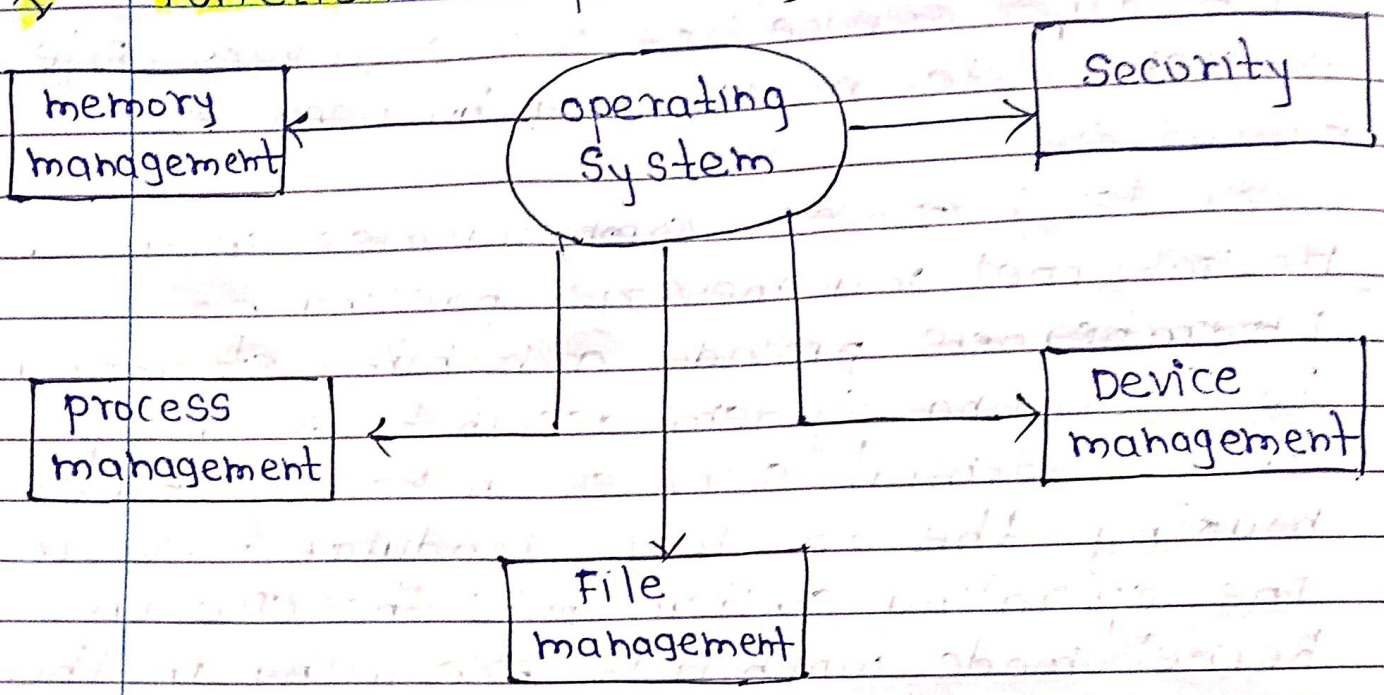
Imp Function of operating System -



- An operating system provides an interface between the user & the hardware.
- It makes a computer more convenient to use.
- In addition, an operating System manages computer resources.
- It allows computer resource to be used in an efficient way.

• process management - process management in operating system is about managing processes.
- A process is a running program.
- operating system makes sure each process:
• gets its turn to use the CPU.
- Synchronized when needed.
• core functions in process management:

• Memory Management -

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- Memory management is an essential task of the operating system that handles the storage & organization of data in both main (primary) memory & secondary storage.
- It also manage the interact between volatile main memory & non-volatile secondary memory.

- File management -

File management in the OS ensures the organized storage, access & control of files.

- It manages how files are stored on different types of storage devices & ensure smooth access through directories & permissions.
- File system management includes managing of File attributes.
File Types.
- Access method

- Device management -

Device management of an operating system handles the communication between the system & its hardware device likes printers, disks or network interfaces.

- Major components in device management
- Device driver
- kernel-space driver
- user space drivers.
- Spooling.

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- Security -

Security mechanism is an operating System are designed to safeguard system resources.

- These mechanism control processes or user can access specific resources & ensure that only authorized users can perform specific actions.

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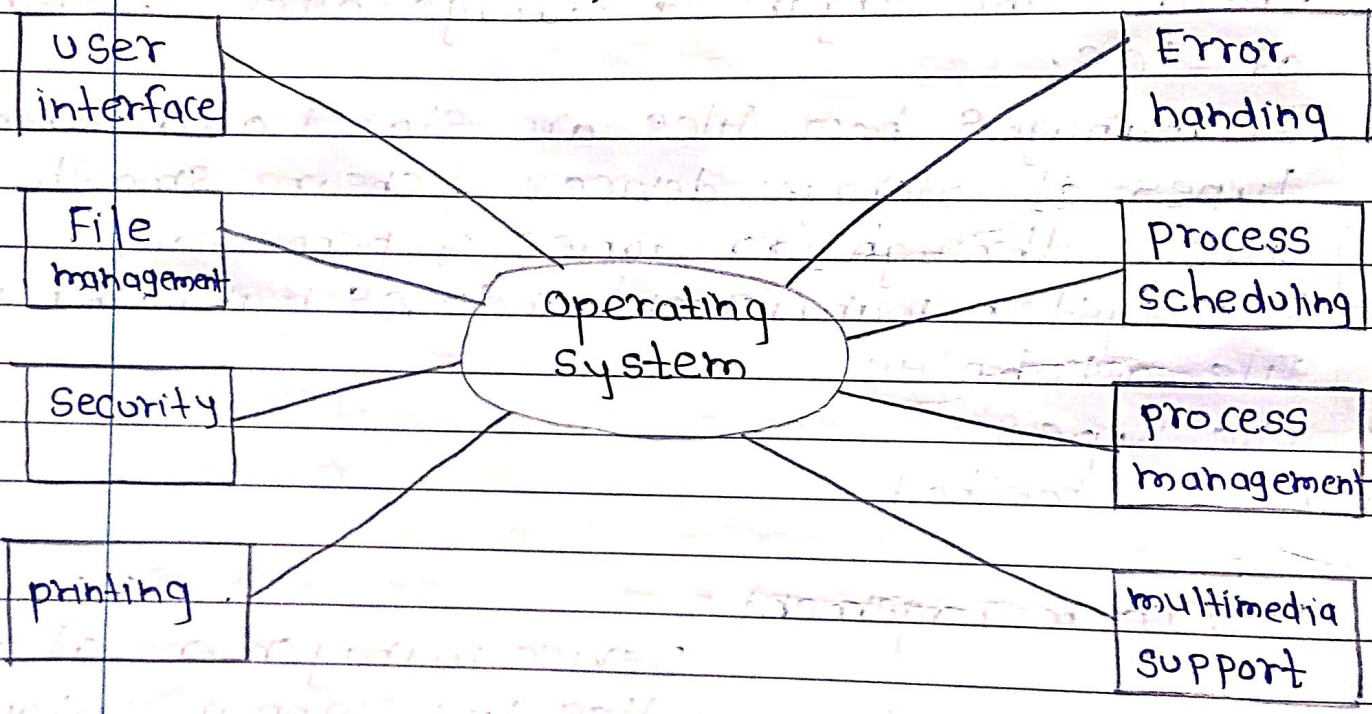
- OS Services:



- operating system services are the core functions that an OS provides to manage a computer's hardware & software resources, enabling user & applications to interact with the system.
- Following are few common service provided by an operating system

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- user interface
- File management
- Security
- printing
- Error Handing
- process Scheduling
- process management
- multimedia support

• user interface-

The os provides a way for users to interact with the computer, either through a graphical user interface (GUI) or a command line interface (CLI).

- This allows user to run programs, manage Files & configure the system.

• File management-

The os organize & manage Files & directories on storage devices.

- It provides Functionalities for creating, deleting, renaming, reading, & writing Files as well as managing permission.

• Security-

The os provides security features to protect the system from unauthorized access & software.

- This includes user authentication, access control & malware.

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- printing-
operating systems manage the printing function by acting between appl & the printer Hardware.
- This allows other system & user tasks to continue while printing occurs in the background.

- Error Handling-
The os detects & handles errors that occurs during program execution or system operation.
- It provides mechanisms for error reporting, debugging & recovery.

- process Scheduling-
The os manages the Computer's execution of multiple program concurrently.
- It handles process creation, termination, scheduling, & communication between processes.

- Multimedia Support-
os services provide essential functionalities that allow users & application to interact with computer H/w.
- Multimedia Support is an aspect of these services, enabling the system to handle & process audio, vedio & other media format.
- For multimedia support the os plays a role in managing resource like audio & vedio devices, handling data streams & providing multimedia hardware

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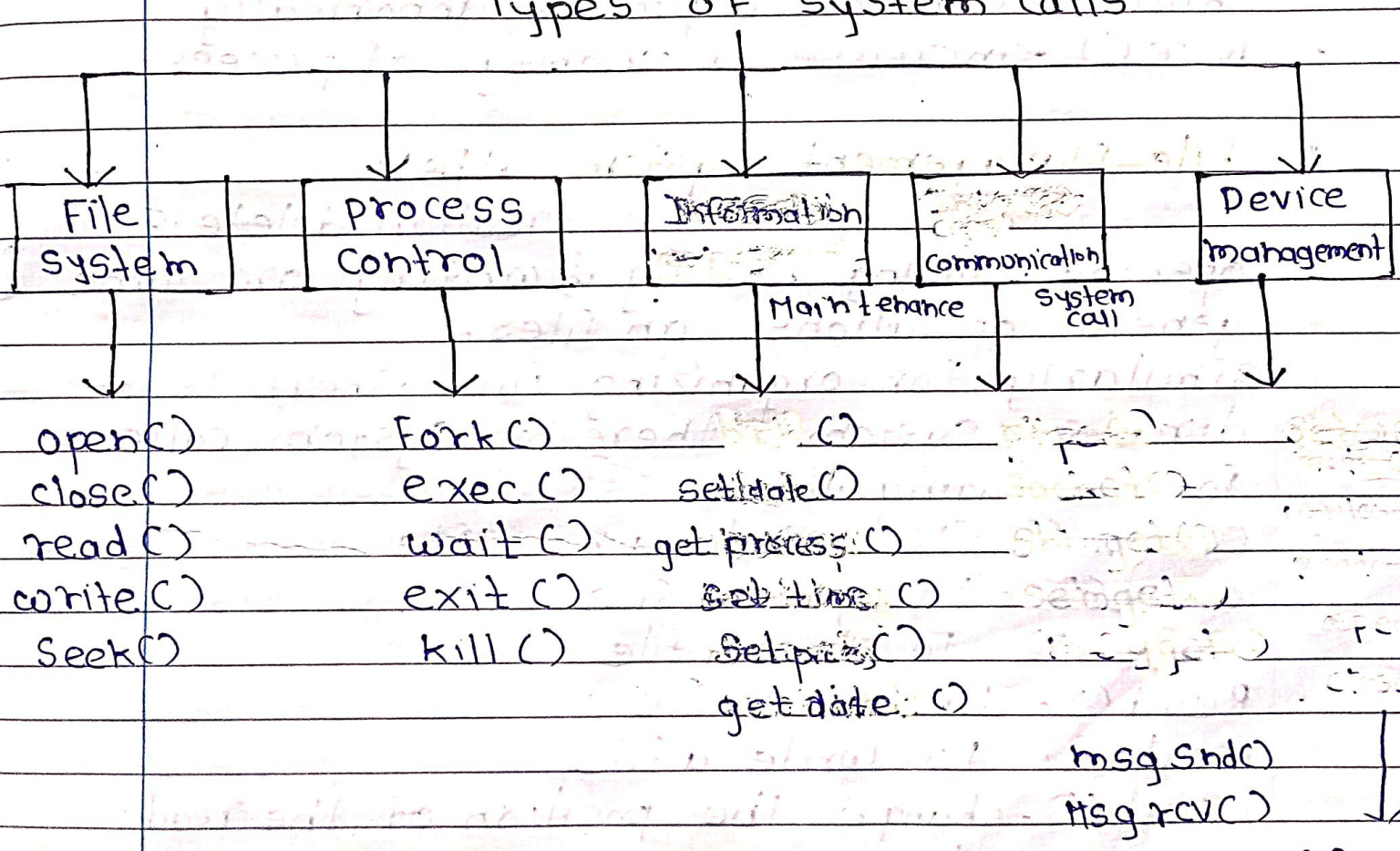
- process management -
The os manages handle the creation, scheduling & termination of processes as well as their synchronization & communication
- The os basically scheduled time for all program to run smoothly & execute all the different program efficiently
- System calls.

Imp. Type of System Calls.



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- **process control** -
  A process a basic entity in the System.
- The processes in the system need to be created, deleted and aborted beside these many operations are required on the processes for their management.
- These are some example.
- Fork() - Create a process
- exist() - Terminate a process
- kill() - Terminate a process abnormally
- nice() - Increase a priority of process.

- **File Management System calls:**
  creation, delete & opening, closing, reading & writing are same general operations on files.
- Similarly for organizing files, there is a directory system & there by System calls for managing them
- Create() - To create a new file
- open() - To open a file
- close() - To close a file
- Read() - To read a file.
- write() - To write a file.
- LSeek() - change the position of the read-write pointer

- **Information maintance System call()** -
  many System calls exits

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Simply for the purpose of transfering information between the user programs & the OS.

- Like → return current time & date
  → no of current users
  → version no of OS.
  → Amount of free memory

get time() get process() get date()
Set date() set process()
get system data()

- **Communication system call()-**

There is a need for communication among the processes in the system.

Msg Snd() - Sending a message
Msg rec() - Receiving a message

- These System call may be related to the communication between processes either the same machine or between processes on different nodes of a network.
- Thus interprocess communication is provided by the OS through these communication related system calls.

- **Device Management()-**

Device management in the context of system calls refers to the operating system's (OS) handing of hardware devices.

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key aspects of Device management of system calls.

- Requesting & Releasing Devices
- Reading & writing

- **System call -**

In operating system, a system call is mechanism that allows a running program (or process) to request services from the operating system.

or

A System call is an interface between a program running in user space & the operating System.

- Introduction to Linux OS.

Linux -

It is an open source operating System kernel.

- It acts as the core of a Computer's software managing hardware resources & enabling communication between hardware & software

Imp. - Components of Linux system:

or

Architecture of Linux operating system.

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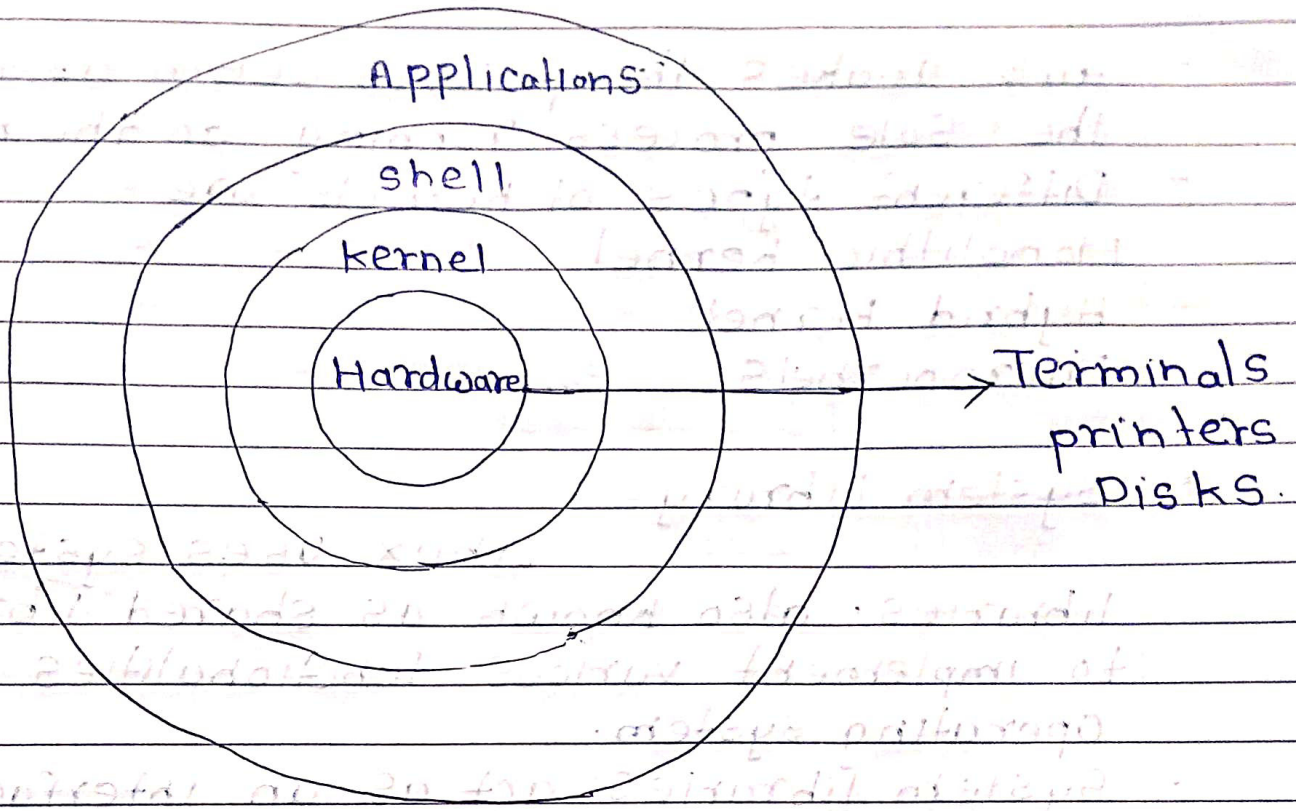

- The Linux operating system can be broken down into 4 key components
The kernel, system libraries, system utilities & the shell.
- The kernel is the core, interacting directly with the hardware.
- System libraries provides functionalities for applications.
- Linux architecture has the following components

• Kernel -
kernel is the core of the Linux based operating system.
- The common hardware resources of the computer to provide each process with it virtual process/resources.

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- This makes the process seem as if it is the sole process running on the machine.
- Different types of kernels are -
- Monolithic kernel
- Hybrid kernel
- Microkernels

- System Library -

Linux uses system libraries, also known as shared libraries, to implement various functionalities of the operating system.

- System libraries act as an interface between applications & the kernel, providing a standard & efficient way for applications to interact with a underlying system.

- Shell -

The shell is the user interface of the Linux operating system.

- It allows users to interacts with the system by entering commands, which the shell executes.
- It provides a convenient way for user to perform various tasks, such as running programs, managing files & the configuring the system.

- Hardware Layer -

The hardware layer

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all the physical components of the computers such as HDD (Hard disk drives), CPU, input/output devices.

- This layer is responsible for interacting with the linux operating system & providing necessary resources for the system

- Application software -
  This is the software that user interact with directly such as web browsers, word processors.